BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 04 CSE470:Software Engineering Course
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Name: (Please write in CAPITAL LETTERS)	ID: Sec:

Answer the following questions.

Which design pattern is used to allow communication between objects in a loosely coupled way? **A. Observer Pattern** B. Singleton Pattern C. Facade Pattern D. Command Pattern	In Singleton Design Pattern, what is the name of the static method that is used to retrieve the instance of the singleton class? **getInstance()**
Which design pattern is used to provide a way to access the elements of an aggregate object sequentially without exposing its underlying representation? A. Adapter Design Pattern B. Facade Design Pattern C. Observer Design Pattern **D. Iterator Design Pattern**	In Observer Design Pattern, which component is responsible for maintaining a list of all the observers and notifying them of any changes in the state? **A. Subject** B. Observer **C. ConcreteSubject** D. ConcreteObserver
In Singleton Design Pattern, what is the name of the static method that is used to retrieve the instance of the singleton class? **getInstance()**	Which design pattern is used to convert the interface of a class into another interface that clients expect?
In Facade Design Pattern, what is the name of the class that provides the simplified interface to the complex subsystem? **A. Facade** B. Subsystem C. Client D. Adapter	Which design pattern allows an object to be notified automatically of changes in another object's state? A. Decorator Design Pattern **B. Observer Design Pattern** C. Command Design Pattern D. Adapter Design Pattern
Which design pattern provides a simplified interface to a complex subsystem? **A. Facade Design Pattern** B. Observer Design Pattern C. Singleton Design Pattern D. Adapter Design Pattern	In Adapter Design Pattern, what is the name of the class that adapts the interface of the adaptee to the target interface? **A. Adapter** B. Adaptee C. Target D. Client

```
interface Observer {
    void update();
}

class Subject {
    private List<Observer> observers = new ArrayList<>();

    void attach(Observer observer) {
        observers.add(observer);
    }

    void detach(Observer observer) {
        observers.remove(observer);
    }

    void notifyObservers() {
        for (Observer observer : observers) {
            observer.update();
        }
    }
}

class ConcreteObserver Implements Observer {
    private String name;

    ConcreteObserver(String name) {
        this.name = name;
    }

    @Override
    void update() {
        System.out.println(name + " received an update.");
    }
}
```

What design pattern is implemented in the code above?

Behavioral/ Observer Design Pattern

What is the role of the Observer interface?

It defines the update method that is called when the subject changes